

Aaron Wu

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EDUCATION

Rutgers University (GPA 4.0) <i>B.S. Computer Science, Data Science</i>	Graduation: 05/2027 or 01/2028 (Flexible)
University of Toronto (GPA 4.0) <i>B.S. Computer Science, Applied Statistics</i>	09/2023 – 04/2024 (Transferred)

EXPERIENCE

Platform Engineer Intern (On-Prem Kubernetes) | *Macronix* 07/2026 – 08/2026

- Overturned a reported **capacity** diagnosis; redesigned dispatch into **independently scalable** per-function services.
- Built **Envoy Gateway** → **Prometheus** → **KEDA** autoscaling on gateway request metrics for single-threaded runtimes.
- Prototyped metric-to-log datalink; sequenced OTel observability roadmap; reprioritized tracing & alert fixing per team needs.
- Traced a month-old **silent failure** to external image pulls; authored the first Helm chart & image SOP, shaped for **ArgoCD**.

AI Solutions Engineering Intern | *Ricoh Company, Ltd.* 06/2026 – 07/2026

- Assessed a partner AI vendor **as a buyer**; turned blanket assurances into **auditable AI governance** commitments.
- Ran AI adoption workshops for **7 cross-functional staffs**; embedded **Claude** into deployment eng. & marketing workflows.

Systems Infrastructure Engineer Intern (RAG Agent Development) | *AMD* 05/2025 – 08/2025

- Rebuilt a codebase-grounded coding agent with **GraphRAG** & LangChain ReAct; **shipped to org-wide agent platform**.
- Engineered **tiered retrieval** over a **6-language** code graph (Neo4j), escalating exact → semantic → graph traversal.
- Diagnosed structurally similar code across classes as the retrieval failure, lifting ideal retrieval rate from **<50% to 85+%**.
- Cut premature answers by tracking **unfinished subtasks** & visited nodes, gating output until exploration completed.

Research and Development Intern | *QT Medical, Inc.* [LINK] 07/2024 – 08/2024

- Cut physician time labeling **ECG** diagnoses **70%** by tracing the bottleneck upstream & building a self-serve platform.
- Owned design to **AWS EC2** deployment in **12 days**; automated data pipelines (S3, RDS) & iterated UX with physicians.

Research Assistant – Randomized Federated Learning | *Rutgers University* 09/2025 – 01/2026

- Conducted empirical analysis for randomized federated hyperparameter-learning algorithm on sparse, **non-IID data**.

PROJECTS

StateFlow – Durable Execution Layer for AI Pipelines [LINK] | *Go, PostgreSQL, Docker*

- Building a **zero-SDK** durable-execution sidecar that resumes crashed pipelines; supports any **HTTP endpoint** as worker.
- Made the database the sole coordination point; **compare-and-set** fences survive crashes so **no step commits twice**.
- Specified every state, transition & failure behavior (including unbuilt features) **before coding**; used **AI as implementer**.

Resilient AI Gateway | *Go, Redis, Lua, Reverse Proxy*

- Architected a reverse-proxy gateway fronting backend LLM deployments, isolating model backends from clients.
- Built a **3-layer rate-limiting** mechanism in **Redis Lua**: model capacity (503), per-IP abuse & per-key quota (429).
- Made degradation explicit with per-layer **fail-open/fail-closed** modes, so a Redis outage cannot disable protection.

FlashForm – High-Concurrency Flash-Sale Platform [LINK] | *Java, Spring Boot, Redis*

- Enforced **exactly-once** order semantics with **Redis Lua** atomic decrements & API-layer idempotency keys; no overselling.
- Held **p99 <50ms** at **2,000+ QPS** on a single containerized node under JMeter load, buffering writes via RabbitMQ.

Real-Time Streaming Data Pipeline (Containerized) | *Kafka, Spark, Airflow, Docker*

- Built a containerized ETL news pipeline: **Kafka** ingestion → **Spark** transformation → schema-driven Cassandra sink.
- Orchestrated the reproducible stack via Docker Compose and **Airflow**, enforcing strict data quality checks at each stage.

OTHER TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, JavaScript, C/C++, SQL, Bash, PowerShell, R, Assembly
Backend & APIs: Spring Boot, Node.js, FastAPI, Flask, REST, GraphQL, gRPC, Microservices, OAuth
Systems & Dist.: Reverse Proxy (Nginx), Event-Driven, TCP/UDP, Low-Latency, High-Throughput, Scalability
AI & Agentic: LangChain, GraphRAG, RAG/RAGAS, Agentic Workflows, Prompting, vLLM, HuggingFace, AI Gateway
Data & ML: Kafka, Spark, Airflow, dbt, Airbyte, ETL, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Power BI
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, Redis, Neo4j (Cypher), Pinecone, ChromaDB
Cloud & DevOps: AWS, Azure, GCP, Docker, Linux/POSIX, CI/CD, GitHub Actions, Prometheus, Grafana, ArgoCD, Scrum
Frontend & Other: React/Native, Next.js, HTML, Figma, Flutter, Streamlit, AI Governance, Code Review, E2E Delivery

CERTIFICATIONS & STUDIES

Certifications: AWS Certified Cloud Practitioner (08/2024); Machine Learning Specialization, Stanford Online (01/2025)
Self-Study: Replication, Partitioning, Consistency, Storage Engines (LSM-Trees, B-Trees), Batch & Stream Processing
Coursework: Systems Programming (Multithreading), Internet Tech (TCP/IP, Sockets), Algorithms, Computer Architecture